

STE Mapbook Workflow — User Guide

This guide walks you through loading, configuring, and running the STE Mapbook Workflow, which generates a multi-section mapbook report for wildlife tracking data sourced from EarthRanger and Google Earth Engine.

What it produces

- Six interactive map visualizations (movement tracks, speed, day/night, home range, mean speed raster, seasonal home range)
- A Word document mapbook (`.docx`) with a cover page and one section per subject
- An interactive widget dashboard

Requirements

- Access to an **EarthRanger** instance with a configured data source
 - Access to a **Google Earth Engine** project
 - The `landDx` geodatabase (downloaded automatically from Dropbox by default — no local copy required)
-

1. Load the Workflow

In the workflow runner, add a new template using this repository's URL:

```
https://github.com/wildlife-dynamics/ste-mapbook.git
```

Once added, select **STE Mapbook Workflow** from the available templates list to load it.

2. Configure the Workflow

You'll be prompted to fill in the following parameters before running.

Workflow Details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional description

Data Source Connections

Both an **EarthRanger** connection and a **Google Earth Engine** connection are required for the workflow to run.

Field	Description
Connect to EarthRanger	Select (or create) the EarthRanger connection to pull observations from
Connect to Earth Engine	Select the Google Earth Engine project used for seasonal analysis

Analysis Time Range

Field	Description
Since	Start date/time of the analysis period
Until	End date/time of the analysis period

Previous Period Range

Defines the comparison period shown on the Movement Tracks map.

Option	Description
Same as current period	Mirrors the current period length, ending at the current start date
Previous month	Calendar month before the current period
Previous 3 months	Three calendar months before the current period
Previous 6 months	Six calendar months before the current period
Previous year	One calendar year before the current period
Enter start date	Manually specify a start date for the previous period

Subject Group

Field	Description	Default
Subject Group Name	Name of the subject group in EarthRanger (case-sensitive)	Elephants

Map Overlay

Choose the overlay layer to display on all maps.

Option	Description	Default
LandDx	Protected area boundaries (community conservancies, national reserves, national parks), downloaded from a <code>.gpkg</code> URL or a local file	Pre-filled Dropbox URL
EarthRanger Spatial Feature	Fetches a named spatial feature set directly from your connected EarthRanger instance	—

By default, **LandDx** is pre-selected with the download URL filled in — no action required unless you want to use an EarthRanger spatial feature instead.

Trajectory Segment Filter

Removes GPS noise and unrealistic movements before trajectory analysis. Applied to both current and previous period trajectories.

Field	Default	Description
Minimum Segment Length (m)	<code>0.001</code>	Discard segments shorter than this distance
Maximum Segment Length (m)	<code>5000</code>	Discard segments longer than this distance
Minimum Segment Duration (s)	<code>1</code>	Discard segments shorter than this duration
Maximum Segment Duration (s)	<code>21600</code>	Discard segments longer than this duration (6 hours)
Minimum Speed (km/h)	<code>0.01</code>	Discard segments below this average speed
Maximum Speed (km/h)	<code>9</code>	Discard segments above this average speed

Adjust these values to suit the movement characteristics of your study species — overly tight bounds can filter out most of a subject's data.

Zoom to GDF Extent

Field	Default	Description
Expansion Factor	1.05	Padding around the map boundary. 1.0 = tight fit, 1.2 = 20% padding

Report Logo

Appears on the mapbook cover page.

Option	Description
Download from URL	Provide a URL to a PNG or JPG logo
Use local file	Provide a path to a local image file

3. Run the Workflow

Once all parameters are configured, submit the workflow. The runner will:

1. Pull movement data from EarthRanger for the specified subject group and time range.
2. Filter trajectories using the segment filter settings.
3. Compute home ranges, speed rasters, and seasonal ranges using Google Earth Engine.
4. Generate all map visualizations and the Word mapbook.
5. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS` .

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/` :

File	Description
<code>trajectories.geoparquet</code>	Current period trajectories
<code>previous_period_trajectories.geoparquet</code>	Previous period trajectories
<code>relocations.geoparquet</code>	Current period relocations
<code>previous_period_relocations.geoparquet</code>	Previous period relocations
<code>*.geoparquet</code> (ETD, MCP, seasonal)	Home range polygons per subject
<code>*_movement_tracks.html</code>	Interactive movement tracks map per subject

File	Description
*_speedmap.html	Interactive speed map per subject
*_day_night.html	Interactive day/night map per subject
*_homerange.html	Interactive home range map per subject
*_mean_speed_raster.html	Interactive mean speed raster map per subject
*_seasonal_home_range.html	Interactive seasonal home range map per subject
mapbook_context_page.docx	Cover page document
*.docx (per subject)	Individual subject report sections
Merged mapbook .docx	Final combined Word report

More Help

- **Full documentation site:** wildlife-dynamics.github.io/ste-mapbook — user guide, technical guide, and troubleshooting
- **Issues:** github.com/wildlife-dynamics/ste-mapbook/issues